

17. (1 point)

Given the following information:

- Premium Development to Loss Development (PDLD) at 3rd retro adjustment: 0.40
- PDLD at 4th retro adjustment: 0.25
- Cumulative loss development factor at 2nd retro adjustment: 1.60
- Cumulative loss development factor at 3rd retro adjustment: 1.25
- Assume all losses are settled at the 4th retro adjustment.

Calculate the cumulative PDLD ratio at the 3rd retro adjustment.

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EXAM 7 SPRING 2017 SAMPLE ANSWERS AND EXAMINER'S REPORT

QUESTION 17

TOTAL POINT VALUE: 1

LEARNING OBJECTIVE(S): A14

SAMPLE ANSWERS

Sample 1

$$\% \text{Loss 2nd} = \frac{1}{1.6} = 0.625$$

$$\% \text{Loss 3rd} = \frac{1}{1.25} = 0.8$$

$$1 - 0.8 = 0.2$$

$$0.8 - 0.625 = 0.175$$

$$\text{CPDLD } 3^{\text{rd}} = \frac{0.4 \times 0.175 + 0.25 \times 0.2}{0.175 + 0.2} = \frac{0.12}{0.375} = 0.32$$

Sample 2

#	PDLD	CDF	% reported	(inc) % emerged
1				
2		1.6	0.625	
3	0.4	1.25	0.8	0.175 = 0.8 – 0.625
4	0.25	1	1	0.2

$$\text{CPDLD}_3 = \frac{0.4 \times 0.175 + 0.25 \times 0.2}{0.175 + 0.2} = 0.32$$

Sample 3

$$\% \text{ of loss reported at } 2^{\text{nd}} \text{ adj.} = \frac{1}{1.6} = 0.625$$

$$\% \text{ of loss reported at } 3^{\text{rd}} \text{ adj.} = \frac{1}{1.25} = 0.8$$

$$\text{Incremental loss reported at } 2^{\text{nd}} - 3^{\text{rd}} \text{ adj.} = 0.175$$

$$\text{Incremental loss reported at } 3^{\text{rd}} - 4^{\text{th}} \text{ adj.} = 0.2$$

$$\text{CPDLD}_3 = \frac{0.4 \times 0.175 + 0.25 \times 0.2}{0.175 + 0.2} = 0.32$$

EXAMINER'S REPORT

Candidates were expected to demonstrate understanding of the connection between loss reporting pattern PDLD and CPDLD ratios in the context of retrospective rating adjustments.

Candidates were expected to know the formula and apply it using the information given. The key steps are identifying the CPDLD formula, calculating the incremental reporting pattern between the retro adjustments, and correctly calculating the result.

Common errors include:

- Using weights other than incremental percent reported in the solution. Some candidates used the cumulative percent reported or the percent unreported.
- Using PDLD ratios weighted with incremental percent reported from the wrong development period.
- Not knowing the CPDLD formula.